8. A simple pendulum has a time period T_1. When its point of suspension is moved vertically upwards according as $y = k\,t^2$, where y is vertical distance covered and $k = 1\ ms^{-2}$, its time period becomes T_2. Then, $\frac{T_1^2}{T_2^2}$ is (g = 10 m s^{-2}) (IIT 2005)

a) $\frac{5}{6}$ b) $\frac{11}{10}$

c) $\frac{6}{5}$ d) $\frac{5}{4}$

9. An ideal spring of spring constant k, is suspended from the ceiling of a room and a block of mass M is fastened to its lower end. If the block is released when the spring is un-stretched, then the maximum extension in the spring is (IIT 2002)

a) $4\frac{Mg}{k}$ b) $\frac{Mg}{k}$

c) $2\frac{Mg}{k}$ d) $\frac{Mg}{2k}$

10. A pendulum is hung in a very high building oscillates to and fro motion freely like a simple harmonic oscillator. If the acceleration of the bob is 16 ms^{-2} at a distance of 4 m from the mean position, then the time period is (NEET 2018 model)

a) 2 s b) 1 s

c) 2πs (d) πs

11. A hollow sphere is filled with water. It is hung by a long thread. As the water flows out of a hole at the bottom, the period of oscillation will

a) first increase and then decrease

b) first decrease and then increase

c) increase continuously

d) decrease continuously

12. The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are (AIPMT 2012)

a) kg m s^{-1} b) kg m s^{-2}

c) kg s^{-1} (d) kg s

13. Let the total energy of a particle executing simple harmonic motion with angular frequency is 1 rad s^{-1} is 0.256 J. If the displacement of the particle at time $t = \frac{\pi}{2}s$ is $8\sqrt{2}$ cm then the amplitude of motion is

a) 8 cm b) 16 cm

c) 32 cm d) 64 cm

14. A particle executes simple harmonic motion and displacement y at time t_0, $2t_0$ and $3t_0$ are A, B and C, respectively. Then the value of $\left(\frac{A+C}{2B}\right)$ is

(a) cos ωt_0 (b) cos $2\omega t_0$

(c) cos $3\omega t_0$ (d) 1

15. A mass of 3 kg is attached at the end of a spring moves with simple harmonic motion on a horizontal frictionless table with time period 2π and with amplitude of 2m, then the maximum fore exerted on the spring is

(a) 1.5 N (b) 3 N

(c) 6 N (d) 12 N

Answers:

1) d	2) c	3) a	4) b
5) b	6) b	7) d	8) c
9) c	10) d	11) a	12) c
13) b	14) a	15) c	

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